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Skin care composition

Abstract

The present invention generally relates to the field of skin care. More particularly, the invention relates to a cosmetic or therapeutic skin care composition comprising live bacteria of at least one *Cutibacterium acnes* (*C. acnes*) strain in combination with a preservative that specifically supports their viability during storage and/or their ability to replicate after application to the skin. Preferably, the cosmetic or therapeutic skin care composition comprises bacteria of at least one *C. acnes* strain selected from the group consisting of D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4. The invention also provides a method for treating or preventing acne by applying the skin care composition of the invention to a skin area in need of treatment. The invention also relates to the use of a skin care composition of the invention for treating or preventing acne.

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Background/Summary

(1) The present invention generally relates to the field of skin care. More particularly, the invention relates to a cosmetic or therapeutic skin care composition comprising live bacteria of at least one *Cutibacterium acnes* (*C. acnes*) strain in combination with a preservative that specifically supports their viability during storage and/or their ability to replicate after application to the skin. Preferably, the cosmetic or therapeutic skin care composition comprises bacteria of at least one *C. acnes* strain selected from the group consisting of D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4. The invention also provides a method for treating or preventing acne by applying the skin care composition of the invention to a skin area in need of treatment. The invention also relates to the use of a skin care composition of the invention for treating or preventing acne.

BACKGROUND OF THE INVENTION

(2) Common acne is a widespread long-term skin condition that affects more than 600 million people worldwide. Acne is most common among teenagers, though it affects people of all ages. It is normally caused by a combination of sebaceous gland hyperplasia, excessive sebum production, and impaired cornification. As a result, the hair follicles become plugged with oil and dead skin cells which leads to pimples and oily skin (Pschyrembel, Klinisches Wörterbuch, 258. ed., Walter de Gruyter-Verlag, Berlin, 1998). The colonization of the affected skin area with bacteria may additionally cause inflammation. Acne predominantly affects skin areas with a high number of sebaceous glands, in particular the face, the upper part of the chest, and the back. The occurrence of acne may lead to emotional distress and mental problems, such as reduced self-esteem and depression.

(3) Oily skin is a transition state between healthy skin and acne-prone skin. In oily skin, the sebaceous glands of the skin produce an excessive amount of sebum which then serves as an ideal nutrient for a number bacteria and yeasts, including the anaerobic gram-positive bacterium *Cutibacterium acnes* (formerly known as *Propionibacterium acnes*) and different species of the yeast genus *Pityrosporum*. These microorganisms decompose the sebum to glycerine and fatty acids, thereby further inducing the production of sebum in the sebaceous glands and destroying the follicle walls in the skin. This results in inflammation of the skin and the formation of pimples, pustules, nodules and cysts which often heal only with scarring which permanently affects the optical appearance of the subject's skin (W. Umbach [Ed.], Kosmetik, Entwicklung, Herstellung und Anwendung kosmetischer Mittel, 2. Edition Thieme Verlag, Stuttgart, 1995).

(4) A number of different factors appear to contribute to the occurrence of acne, including genetics,

hormone status, stress and diet. In addition, the anaerobic bacterial species *Cutibacterium acnes* (formerly *Propionibacterium acnes*) is thought to play an important role in the development of acne, since high numbers of these bacteria are regularly found in patients suffering from moderate or severe inflammatory acne. However, the underlying mechanisms are not completely understood. (5) Today, the standard treatment for acne normally includes the topical application of antibiotics, including erythromycin, clindamycin, metronidazole, sulfacetamide, doxycycline or minocycline, to reduce the number of bacteria, especially *C. acnes*. However, some of these antibiotics exhibit considerable side effects which make their use inconvenient for the patient. In addition, the treatment of acne with antibiotics is associated with high relapse rates due to the fact that small populations of *C. acnes* survive and resume growth after termination of the antibiotic treatment. Accordingly, there is a need for new methods of treating or preventing acne which is essentially free of side effects and provides for a long-lasting effect.

(6) In more recent treatment approaches, acne has been considered to be the result of a distortion of the human skin microbiome caused by specific strains of *C. acnes* (Holmes, 2013; Lomholt and Kilian, 2010). Researchers only recently began to investigate the skin microbiome (Belkaid and Segre, 2014; Oh et al., 2014). While the skin is colonized by a large number of microorganisms which are harmless or even beneficial (Grice and Segre, 2011), alterations of the microbiome can result in diseases such as acne (Bek-Thomsen et al., 2008; Holmes, 2013; Kong et al., 2012, Fitz-Gibbon et al., 2013). This distortion is likely caused by a specific subset of the skin bacterium *C. acnes* (Lomholt and Kilian, 2010). It has therefore been suggested to modulate the skin microbiome in the attempt to restore a healthy microbiome. For example, WO 2016/172196 A1 discloses a method of treating acne in a subject by first administering a disinfectant or antibiotic and subsequently administering a composition comprising one or more live *C. acnes* strain to the skin of the subject. Similarly, WO 2018/073651 A1 discloses a composition for acne treatment comprising two or more different *C. acnes* strains, including *C. acnes* strain C3 and/or K8.

SUMMARY OF THE INVENTION

(7) In a first aspect, the invention relates to a skin care composition for topical administration to the skin comprising (a) lyophilized or spray-dried live bacteria of at least one *Cutibacterium acnes* (*C. acnes*) strain; and (b) a preservative selected from the group consisting of ethanol, phenoxyethanol, caprylyl glycol, methylpropanediol, and mixtures thereof.

(8) In some embodiments, the skin care composition comprises ethanol as a preservative. In some embodiments, ethanol is present in the skin care composition in an amount of 0.5 to 20.0% (w/w), more preferably 5.0 to 15.0% (w/w).

(9) In some embodiments, the skin care composition comprises phenoxyethanol as a preservative. In some embodiments, phenoxyethanol is present in the skin care composition in an amount of 0.05 to 0.5% (w/w), more preferably 0.1 to 0.2% (w/w).

(10) In some embodiments, the skin care composition comprises caprylyl glycol as a preservative. In some embodiments, caprylyl glycol is present in the skin care composition in an amount of 0.05 to 0.5% (w/w), more preferably 0.1 to 0.2% (w/w).

(11) In some embodiments, the skin care composition comprises methylpropanediol as a preservative. In some embodiments, methylpropanediol is present in the skin care composition in an amount of 0.05 to 5.0% (w/w), more preferably 1.0 to 5.0% (w/w).

(12) In some embodiments, the skin care composition comprises a mixture of phenoxyethanol, caprylyl glycol and methylpropanediol as a preservative.

(13) In some embodiments, the skin care composition does not comprise any other preservative except for ethanol, phenoxyethanol, caprylyl glycol, methylpropanediol.

(14) In some embodiments, the skin care composition further comprises an emollient selected from the group consisting of dicaprylyl carbonate, ethylhexyl cocoate, and mixtures thereof.

(15) In some embodiments, the skin care composition further comprises a filler selected from the group consisting of distarch phosphate, tapioca starch, and mixtures thereof.

- (16) In some embodiments, the skin care composition further comprises a thickener selected from the group consisting of a *Chondrus crispus* extract, hydroxypropyl starch phosphate, and mixtures thereof.
- (17) In some embodiments, the skin care composition further comprises PEG-40 hydrogenated castor oil as a solubilizer.
- (18) In some embodiments, the skin care composition further comprises citric acid/citrate buffer as a pH adjuster.
- (19) In some embodiments, the skin care composition comprises at least one *C. acnes* strain selected from the group consisting of single locus sequence typing (SLST) type strains D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4. In some embodiments, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain. In some embodiments, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type K8 strain. In some embodiments, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain.
- (20) In some embodiments, the skin care composition further comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type A5 strain. In some embodiments, the skin care composition further comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type F4 strain.
- (21) In some embodiments, the concentration of each *C. acnes* strain is at least 0.5% (w/v) of the skin care composition. In some embodiments, the at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain are at approximately equal concentrations within the composition. In some embodiments, the at least one *C. acnes* SLST type C3 strain is present at a higher concentration than said at least one *C. acnes* SLST type K8 strain within the composition. In other embodiments, the at least one *C. acnes* SLST type K8 strain is present at a higher concentration than said at least one *C. acnes* SLST type C3 strain within the composition.
- (22) In some embodiments, each of the *C. acnes* strains in the composition is present in an amount of 10^{4-11} colony forming units per ml (CFU/ml), preferably 10^{7-10} CFU/ml. In some embodiments, the overall amount of bacteria in the composition is 10^{4-11} CFU/ml, preferably 10^{7-10} CFU/ml.
- (23) In some embodiments, the skin care composition is in the form of a gel, cream, ointment or lotion.
- (24) In a second aspect, the invention relates to a method of improving the appearance of the skin of a subject and/or modulating the sebum production of skin cells of a subject and/or maintaining healthy skin of a subject, said method comprising the topical administration of a skin care composition of the invention. In some embodiments, the subject is a human.
- (25) In a third aspect, the invention relates to a method of treating or preventing a condition selected from the group consisting of acne, oily skin, progressive macular hypomelanosis, dandruff, atopic eczema, atopic dermatitis and rosacea in a subject, said method comprising the topical administration of a skin care composition of the invention. In some embodiments, the subject is a human.

Description

BRIEF DESCRIPTION OF THE FIGURE

- (1) FIG. 1 shows the composition of prototype formulations 1-15 used in the reactivation experiments with freeze-dried bacteria.
- (2) FIG. 2 shows the results from experiments analyzing the reactivation of freeze-dried bacteria from prototype formulations 1-11.

(3) FIG. 3 shows the results from experiments analyzing the reactivation of freeze-dried bacteria from prototype formulations 12-15.

(4) FIG. 4 shows the composition of additional prototype formulations 1-11 used in reactivation experiments with freeze-dried bacteria.

(5) FIG. 5 shows the results from experiments analyzing the reactivation of freeze-dried bacteria from the additional prototype formulations 1-11.

(6) FIG. 6 shows the results from experiments analyzing the reactivation of freeze-dried bacteria from comparative formulations V1-V10.

(7) FIG. 7 shows the composition of comparative formulations V1-V10 used in reactivation experiments with freeze-dried bacteria.

DETAILED DESCRIPTION

(8) While these method and compositions provide for a novel and highly effective modulation of the microbiome which ameliorates or eliminates the symptoms of acne, the provision of live bacteria in cosmetic skin care compositions has been found to be associated with significant problems. For example, some of the components commonly used in commercial skin care products interfere with bacterial viability. When these compounds are combined into compositions with lyophilized or spray-dried bacteria, they can either interfere with the viability of the bacteria or negatively influence their growth rate after application to the skin. Accordingly, there is a need for novel skin care products which have been formulated in a way to be compatible with the application of lyophilized or spray-dried bacteria and which do not inhibit the outgrowth of these bacteria after application of the product to a skin.

(9) It has now been surprisingly found that the viability of lyophilized or spray-dried bacteria, in particular bacterial strains of the species *C. acnes*, during storage in the skin care composition as well as their ability to grow upon application to skin can be preserved or significantly improved by the excipients which are added to the formulation of the final skin care product. The present invention is based on the identification of a group of excipients for use in skin care compositions which do not significantly interfere with or may improve bacterial viability and reactivation of growth. These compounds have been found to be highly compatible the administration of bacteria, in particular bacterial strains of the species *C. acnes*.

(10) Thus, in a first aspect the present invention relates to a skin care composition for topical administration to the skin, said composition comprising, consisting essentially of, or consisting of (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain; and (b) a preservative selected from the group consisting of ethanol, phenoxyethanol, caprylyl glycol, methylpropanediol, and mixtures thereof.

(11) In the context with the present invention, lyophilized or spray-dried live bacteria are used in the skin care composition. This means that viable bacteria have been subjected to a drying process that maintains their viability, but reduces their metabolic processes to minimum. In lyophilized or spray-dried form, the bacteria can be stored for months or even years. Once they are applied to the skin, such as the human skin, the metabolism of the bacteria is reactivated such that they resume growth. They propagate on the skin surface and displace pathogenic bacterial strains, thereby recovering a diverse, healthy and balanced skin microbiome.

(12) In one embodiment, the live *C. acnes* bacteria are present in spray-dried form. The principle of spray drying is based on the dispersion of a solution into fine droplets which are introduced into a flow of hot air. The solvent evaporates from the substrate droplets so that dry product clusters remain. Standard spray drying devices can be used, such as the Mini Spray Dryer B-290 from Buchi Labortechnik GmbH (Essen, Germany) or the Mobile Minor™ Spray Dryer from GEA (Berlin, Germany).

(13) In one embodiment, the live *C. acnes* bacteria are present in freeze-dried or lyophilized form. Freeze drying or lyophilization is a process which includes freezing the product, reducing the pressure and adding heat to allow the frozen water in the material to sublime. Various methods

can be applied for freezing the product. For example, freezing can be achieved by using a standard freezer or a chilled bath. Cooling the product below its triple point ensures that sublimation will occur upon heating. To prevent the formation of large crystals that may damage the structure of the product to be dried, freezing is done rapidly. About 95% of the water in the product is removed when the frozen water sublimates. Most materials can be dried to 1-5% residual moisture. Standard freeze drying devices can be used, such as the Lyovac™ devices from GEA (Berlin, Germany), the Gamma 2-20 Freeze dryer LCM-1 from Christ (Osterode am Harz, Germany), or the Christ Martin™ Alpha 1-2 Lyophilisator from Fisher Scientific GmbH (Schwerte, Germany).

(14) The one or more excipients that do not interfere with viability and reactivation of growth of said lyophilized or spray-dried live bacteria are selected from the group consisting of emollients, thickeners, fillers, antioxidants, preservatives, solubilizers, and pH adjusters. Suitable members for each of these groups of compounds will be discussed herein below.

(15) Preservatives

(16) According to the present invention, the skin care composition comprises a preservative. As used herein, a preservative is a compound that is added to cosmetic formulation to prevent microbial spoilage of the formulation by inhibiting the growth of unintended bacteria and yeasts. Commonly used preservatives for cosmetic formulations include, amongst others, benzyl alcohol, salicylic acid and sorbic acid.

(17) According to the present invention, the skin care composition comprises the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and a preservative selected from the group consisting of ethanol, phenoxyethanol, caprylyl glycol, methylpropanediol, and mixtures thereof. These preservatives are surprisingly tolerated to some extent by *C. acnes* strains while being effective against other bacteria that may contaminate the composition.

(18) In one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and ethanol as a preservative. In another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and phenoxyethanol as a preservative. In yet another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and caprylyl glycol as a preservative. In yet another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and methylpropanediol as a preservative.

(19) The skin care composition of the present invention may furthermore comprise more than one preservative, e.g. two or more of the above mentioned preservatives. For example, the composition may comprise ethanol in combination with phenoxyethanol as preservatives. Alternatively, the composition may comprise ethanol in combination with caprylyl glycol as preservatives. Alternatively, the composition may comprise ethanol in combination with methylpropanediol as preservatives. The composition may also comprise phenoxyethanol in combination with caprylyl glycol as preservatives or a combination of phenoxyethanol in combination with methylpropanediol as preservatives. The composition may also comprise caprylyl glycol in combination with methylpropanediol as preservatives.

(20) The best results in terms of protecting the skin care composition of the invention from spoilage with other bacteria or fungi and, at the same time, maintain the viability of the lyophilized or spray-dried live *C. acnes* bacteria in the composition are obtained with a combination of phenoxyethanol, caprylyl glycol and methylpropanediol. Hence, in a particular preferred embodiment, the skin care composition of the present invention comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and a combination of phenoxyethanol, caprylyl glycol and methylpropanediol. For example, when using a combination of phenoxyethanol, caprylyl glycol and methylpropanediol, phenoxyethanol is preferably used in an amount of 0.1 to 0.3% (w/w), more preferably 0.15 to 0.25% (w/w), and more preferably 0.23% (w/w), caprylyl glycol is preferably used in an amount of 0.1 to 0.3% (w/w), more preferably 0.15 to 0.25% (w/w), and more preferably

0.18% (w/w), and methylpropanediol is preferably used in an amount of 2.0 to 5.0% (w/w), more preferably 3.0 to 4.5% (w/w), and more preferably 4.0% (w/w). A skin care composition comprising 0.23% (w/w) phenoxyethanol, 0.18% (w/w) caprylyl glycol, and 4.0% (w/w) methylpropanediol is particularly preferred herein.

(21) When ethanol is used as a preservative, it is preferably used in the final skin care composition in an amount of 0.5 to 20.0% (w/w), more preferably 8.0 to 15.0% (w/w), and more preferably 5.0 to 15.0% (w/w). Stated differently, the amount of ethanol in the skin care composition of the invention may be at least 0.5% (w/w), at least 1.0% (w/w), at least 2.0% (w/w), at least 3.0% (w/w), at least 4.0% (w/w), at least 5.0% (w/w), at least 6.0% (w/w), at least 7.0% (w/w), at least 8.0% (w/w), at least 9.0% (w/w), or at least 10.0% (w/w).

(22) When phenoxyethanol is used as a preservative, it is preferably used in the final skin care composition in an amount of 0.05 to 0.5% (w/w), more preferably 0.1 to 0.25% (w/w), and more preferably 0.1 to 0.2% (w/w). Stated differently, the amount of phenoxyethanol in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.075% (w/w), at least 0.1% (w/w), at least 0.15% (w/w), at least 0.2% (w/w), or at least 0.25% (w/w). The same amounts apply when using caprylyl glycol as a preservative.

(23) When methylpropanediol is used as a preservative, it is preferably used in the final skin care composition in an amount of 0.05 to 5.0% (w/w), more preferably 0.5 to 2.5% (w/w), and more preferably 1.0 to 1.5% (w/w). Stated differently, the amount of methylpropanediol ethanol in the skin care composition of the invention may be at least 0.5% (w/w), at least 1.0% (w/w), at least 2.0% (w/w), at least 3.0% (w/w), or at least 4.0% (w/w), or at least 5.0% (w/w).

(24) When two or more of the above preservatives are used in combination with each other, it is preferred that the overall amount of preservative does not exceed 5.0% (w/w) when using a combination of phenoxyethanol and caprylyl glycol. In particular, when using caprylyl glycol, phenoxyethanol, and methylpropanediol in combination, it is preferred to use these compounds in a ratio of 1:2:20, for example, 0.05% (w/w) caprylyl glycol, 0.1% (w/w) phenoxyethanol, and 1.0% (w/w) methylpropanediol, or alternatively, 0.1% (w/w) caprylyl glycol, 0.2% (w/w) phenoxyethanol, and 2.0% (w/w) methylpropanediol.

(25) The above-mentioned preservatives can be used in combination with other preservatives, in particular short-chain fatty acids such as formic acid, propionic acid or isobutyric acid. In addition, the compositions of the invention may also comprise lactic acid as an additional preservative.

(26) Emollient

(27) In a preferred aspect, the skin care composition further comprises an emollient. As used herein, an emollient is a compound that moisturizes and/or softens the skin. Emollients normally reduce the roughness, cracking and/or irritation of the skin by penetrating into the deeper layers of the skin. Emollients commonly used in skin care products comprise plant oils, like sesame oil, coconut oil, olive oil, almond oil, macadamia nut oil, cottonseed oil or peanut oil, silicone oils, like dimethylpolysiloxane and cyclomethicone, fatty acids, and fatty alcohol ethers.

(28) In a preferred embodiment, the skin care composition comprising the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and the above-defined preservative comprises an emollient selected from the group consisting of dicaprylyl carbonate, ethylhexyl cocoate, and mixtures thereof. In one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and dicaprylyl carbonate as an emollient. In another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and ethylhexyl cocoate as an emollient. In yet another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and both dicaprylyl carbonate and ethylhexyl cocoate as emollients. It is particularly preferred that the skin care composition does not comprise any other emollient except for dicaprylyl carbonate and/or ethylhexyl cocoate.

(29) When dicaprylyl carbonate is used as an emollient, it is preferably used in the final skin care composition in an amount of 0.05 to 25.0% (w/w), more preferably 2.0 to 20.0% (w/w), and more preferably 5.0 to 10.0% (w/w) or 7.5 to 10.0% (w/w). Stated differently, the amount of dicaprylyl carbonate in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.1% (w/w), at least 0.25% (w/w), at least 0.5% (w/w), at least 0.75% (w/w), at least 1.0% (w/w), at least 1.25% (w/w), at least 1.5% (w/w), at least 1.75% (w/w), at least 2.0% (w/w), at least 2.5% (w/w), at least 3.0% (w/w), at least 4.0% (w/w), at least 5.0% (w/w), at least 6.0% (w/w), at least 7.0% (w/w), at least 8.0% (w/w), or at least 9.0% (w/w).

(30) When ethylhexyl cocoate is used as an emollient, it is preferably used in the final skin care composition in an amount of 0.05 to 25.0% (w/w), more preferably 1.0 to 10.0% (w/w), and more preferably 5.0 to 10.0% (w/w), or 7.5 to 10.0% (w/w). Stated differently, the amount of ethylhexyl cocoate in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.1% (w/w), at least 0.25% (w/w), at least 0.5% (w/w), at least 0.75% (w/w), at least 1.0% (w/w), at least 1.25% (w/w), at least 1.5% (w/w), at least 1.75% (w/w), at least 2.0% (w/w), at least 2.5% (w/w), at least 3.0% (w/w), at least 4.0% (w/w), at least 5.0% (w/w), at least 6.0% (w/w), at least 7.0% (w/w), at least 8.0% (w/w), or at least 9.0% (w/w).

(31) When dicaprylyl carbonate and ethylhexyl cocoate are used in combination with each other as emollients, it is preferred that the overall amount of emollient is at least 0.05% (w/w), but does not exceed 20.0% (w/w), more preferably does not exceed 15.0% (w/w) or 10.0% (w/w). In such an embodiment, it is furthermore preferred that dicaprylyl carbonate and ethylhexyl cocoate are used in equal amounts, for example, 2.0% (w/w) dicaprylyl carbonate in combination with 2.0% (w/w) ethylhexyl cocoate, or 5.0% (w/w) dicaprylyl carbonate in combination with 5.0% (w/w) ethylhexyl cocoate.

(32) Thickener

(33) In another preferred aspect, the skin care composition further comprises a thickener.

Thickeners are compounds that increase the viscosity of a cosmetic or pharmaceutical formulation. Thickeners are often polymers that absorb water and swell up, thereby making the composition more viscous. Thickeners commonly used in skin care products comprise bean gum, xanthan gum, gelatin, Carnuba wax, and stearic acid.

(34) According to the present invention, the skin care composition comprising the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and the preservative comprises a thickener selected from the group consisting of a *C. crispus* extract, hydroxypropyl starch phosphate, and mixtures thereof. In one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and a *C. crispus* extract as a thickener. In another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and hydroxypropyl starch phosphate as a thickener.

(35) When a *C. crispus* extract is used as a thickener, it is preferably used in the final skin care composition in an amount of 0.05 to 7.5% (w/w), more preferably 0.1 to 5.0% (w/w), and even more preferably 0.2 to 4.0% (w/w), 0.2 to 2.0% (w/w), or 0.2 to 1.5% (w/w). Stated differently, the amount of the *C. crispus* extract in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.1% (w/w), at least 0.25% (w/w), at least 0.5% (w/w), at least 0.75% (w/w), at least 1.0% (w/w), at least 1.25% (w/w), at least 1.5% (w/w), at least 1.75% (w/w), at least 2.0% (w/w), at least 2.5% (w/w), at least 3.0% (w/w), at least 4.0% (w/w), or at least 5.0% (w/w).

(36) When hydroxypropyl starch phosphate is used as a thickener, it is preferably used in the final skin care composition in an amount of 0.05 to 10.0% (w/w), more preferably 0.1 to 10.0% (w/w), and even more preferably 0.5 to 7.5% (w/w), 1.0 to 5.0% (w/w), or 1.0 to 2.0% (w/w). Stated differently, the amount of hydroxypropyl starch phosphate in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.1% (w/w), at least 0.25% (w/w), at least 0.5% (w/w), at least 0.75% (w/w), at least 1.0% (w/w), at least 1.25% (w/w), at least 1.5% (w/w), or at

least 1.75% (w/w).

(37) It should be understood that the skin care composition of the present invention may also comprise a combination of a *C. crispus* extract and hydroxypropyl starch phosphate as thickeners. When the *C. crispus* extract and hydroxypropyl starch phosphate are used in combination with each other as thickeners, it is preferred that the overall amount of thickeners is at least 0.05% (w/w), but does not exceed 10.0% (w/w), more preferably does not exceed 5.0% (w/w). In such an embodiment, it is furthermore preferred that two thickeners are used in equal amounts, for example, 2.0% (w/w) *C. crispus* extract and 2.0% (w/w) hydroxypropyl starch phosphate.

(38) pH Adjuster

(39) In another preferred aspect, the skin care composition of the present invention further comprises a pH adjuster. Since the composition of the invention is used on the human skin, it will preferably have a neutral or slightly acidic pH to make it more compatible with the acidic environment of the skin. The composition may have a pH in the range from about 2.5 to about 7.5, preferably from about 4.0 to about 7.0, and more preferably from about 6.0 to about 7.0. While an acidic pH in a cosmetic formulation is normally achieved by adding an acid, such as formic acid, acetic acid, butyric acid, valeric acid, caproic acid, enanthic acid, or caprylic acid, it has been found that these acids could compromise the ability of the bacteria in the composition to grow and replicate after administration to the skin. According to the invention, the pH adjuster preferably is a citric acid/citrate buffer.

(40) Therefore, in another preferred aspect, the skin care composition comprising the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and the preservative also comprises a citric acid/citrate buffer, such as for example a citric acid/sodium citrate buffer, as a pH adjuster. In a preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and a buffer consisting of citric acid and sodium citrate. It is particularly preferred that the skin care composition does not comprise any other pH adjuster except for citric acid/citrate.

(41) When citric acid is used as a pH adjuster, it is preferably used in the final skin care composition in an amount of 0.01 to 1.5% (w/w), more preferably 0.01 to 0.25% (w/w), and even more preferably 0.01 to 0.1% (w/w). Stated differently, the amount of citric acid in the skin care composition of the invention may be at least 0.01% (w/w), at least 0.02% (w/w), at least 0.03% (w/w), at least 0.04% (w/w), at least 0.05% (w/w), at least 0.06% (w/w), at least 0.07% (w/w), at least 0.08% (w/w), at least 0.09% (w/w), or at least 0.1% (w/w).

(42) When sodium citrate is used as a pH adjuster, it is preferably used in the final skin care composition in an amount of 0.01 to 1.5% (w/w), more preferably 0.05 to 1.0% (w/w), and more preferably 0.05 to 0.5% (w/w) or 0.1 to 0.2% (w/w). Stated differently, the amount of sodium citrate in the skin care composition of the invention may be at least 0.01% (w/w), at least 0.02% (w/w), at least 0.03% (w/w), at least 0.04% (w/w), at least 0.05% (w/w), at least 0.06% (w/w), at least 0.07% (w/w), at least 0.08% (w/w), at least 0.09% (w/w), at least 0.1% (w/w), 0.11% (w/w), 0.12% (w/w), 0.13% (w/w), 0.14% (w/w), 0.15% (w/w), 0.16% (w/w), 0.17% (w/w), 0.18% (w/w), 0.19% (w/w), 0.2% (w/w), at least 0.3% (w/w), at least 0.4% (w/w), at least 0.5% (w/w), at least 0.6% (w/w), at least 0.7% (w/w), at least 0.8% (w/w), at least 0.9% (w/w), or at least 1.0%.

(43) When citric acid and sodium citrate are used in combination with each other as pH adjusters, it is preferred that the overall amount of the pH adjusters in the final skin care composition does not exceed 0.3% (w/w). In such an embodiment, it is furthermore preferred that citric acid and sodium citrate are used in a ratio of 1:2, for example, 0.05% (w/w) citric acid and 0.10% (w/w) sodium citrate, or 0.1% (w/w) citric acid in combination with 0.2% (w/w) sodium citrate.

(44) It is particularly preferred that citric acid and the citrate are used in the skin care composition of the invention in amounts which ensure a pH of between 5.0 and 7.0 of the overall composition.

(45) Filler

(46) In yet another preferred aspect, the skin care composition further comprises a filler. As used

herein, a filler is a compound that aids in making the skin care composition more homogeneous by uniformly dispersing in the composition. Fillers are used to improve the sensory properties of the skin. Depending on the filler material, the end product may confer a silky, dry, smooth, or powdery skin feel.

(47) According to another preferred aspect of the present invention, the skin care composition comprising the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and the preservative comprises a filler selected from the group consisting of distarch phosphate, tapioca starch, and mixtures thereof. In one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and distarch phosphate as a filler. In another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and tapioca starch as a filler. In yet another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and both distarch phosphate and tapioca starch as fillers. It is preferred that the skin care composition does not comprise any other filler except for distarch phosphate and/or tapioca starch.

(48) When distarch phosphate is used as filler, it is preferably used in the final skin care composition in an amount of 0.05 to 5.0% (w/w), more preferably 1.0 to 5.0% (w/w), and more preferably 2.0 to 5.0% (w/w) or 2.0 to 3.0% (w/w). Stated differently, the amount of distarch phosphate in the skin care composition of the invention may be at least 0.05% (w/w), at least 0.1% (w/w), at least 0.25% (w/w), at least 0.5% (w/w), at least 0.75% (w/w), at least 1.0% (w/w), at least 1.25% (w/w), at least 1.5% (w/w), at least 1.75% (w/w), at least 2.0% (w/w), at least 2.5% (w/w), at least 3.0% (w/w), or at least 4.0% (w/w). The same amounts apply when using tapioca starch as filler.

(49) When distarch phosphate and tapioca starch are used in combination with each other as fillers, it is preferred that the overall amount of the fillers is at least 0.05% (w/w), but does not exceed 5.0% (w/w), more preferably does not exceed 3.0% (w/w). In such an embodiment, it is furthermore preferred that distarch phosphate and tapioca starch are used in equal amounts, for example, 2.5% (w/w) distarch phosphate in combination with 2.5% (w/w) tapioca starch, or 1.0% (w/w) distarch phosphate in combination with 1.0% (w/w) tapioca starch.

(50) Solubilizer

(51) In another preferred aspect, the skin care composition further comprises a solubilizer. As used herein, a solubilizer is a compound that aids in the solubilization of hydrophobic substances in aqueous and alcoholic formulations. For example, a solubilizer may render feasible the solubilization of perfume oils and other hydrophobic substances, such as vitamins, into an aqueous skin care composition. It has been found herein that polyethylene glycol (PEG)-40 hydrogenated castor oil does not interfere with viability and replication capability of the lyophilized or spray-dried bacteria.

(52) Accordingly, in one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and PEG-40 hydrogenated castor oil. In an even more preferred embodiment, PEG-40 hydrogenated castor oil is the only solubilizer included in the composition.

(53) When PEG-40 hydrogenated castor oil is used as solubilizer, it is preferably used in the final skin care composition in an amount of 0.01 to 2.5% (w/w), more preferably 0.05 to 1.5% (w/w), and more preferably 0.5 to 1.0% (w/w). Stated differently, the amount of PEG-40 hydrogenated castor oil in the skin care composition of the invention may be at least 0.01% (w/w), at least 0.05% (w/w), at least 0.1% (w/w), at least 0.2% (w/w), at least 0.3% (w/w), at least 0.4% (w/w), at least 0.5% (w/w), at least 0.6% (w/w), at least 0.7% (w/w), at least 0.8% (w/w), at least 0.9% (w/w), or at least 1.0% (w/w).

(54) Antioxidant

(55) In another preferred aspect, the skin care composition further comprises an antioxidant. These compounds are normally added to cosmetic formulations to prevent oxidative reactions catalyzed by oxygen radicals that would otherwise result in the decomposition of ingredients in the composition, such as proteins, sugars, and lipids. Antioxidants which are commonly used in cosmetic product include chemicals like butylated hydroxytoluene and butylated hydroxyanisole, as well as plant derived polyphenols, flavonoids, flavanols, stilbens, and terpenes.

(56) In a preferred embodiment, the skin care composition comprises the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and an antioxidant selected from the group consisting of tocopherol, tocopheryl acetate, and mixtures thereof. In one preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and tocopherol as an antioxidant. In another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and tocopheryl acetate as an antioxidant. In yet another preferred embodiment, the skin care composition comprises lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, a preservative as defined above and both tocopherol and tocopheryl acetate as antioxidants. It is particularly preferred that the skin care composition does not comprise any other antioxidant except for tocopherol and/or tocopheryl acetate.

(57) When tocopherol, tocopheryl acetate, or mixtures thereof are used as antioxidants, they can be added in lyophilized or spray-dried form to the lyophilized or spray-dried live bacteria. For example, if the composition of the invention is provided in a form that requires the mixing of the bacteria with a cosmetic or pharmaceutical preparation prior to use, the lyophilized or spray-dried antioxidants may be added to the lyophilized or spray-dried bacteria and stored until reconstitution with the cosmetic or pharmaceutical preparation until use. Where tocopherol, tocopheryl acetate, or mixtures thereof are used in dried form, dried vitamin C may also be added.

(58) When tocopherol is used as an antioxidant, it is preferably used in the final skin care composition in an amount of 0.01 to 2.0% (w/w), more preferably 0.05 to 1.5% (w/w), and more preferably 0.1 to 1.0% (w/w). Stated differently, the amount of tocopherol in the skin care composition of the invention may be at least 0.01% (w/w), at least 0.025% (w/w), at least 0.05% (w/w), at least 0.075% (w/w), at least 0.1% (w/w), at least 0.2% (w/w), at least 0.3% (w/w), at least 0.4% (w/w), at least 0.5% (w/w), at least 0.6% (w/w), at least 0.7% (w/w), at least 0.8% (w/w), at least 0.9% (w/w), or at least 1.0% (w/w). The same amounts apply when using tocopheryl acetate as antioxidant.

(59) When tocopherol and tocopheryl acetate are used in combination with each other as antioxidants, it is preferred that the overall amount of antioxidants is at least 0.01% (w/w), but does not exceed 2.0% (w/w). In such an embodiment, it is furthermore preferred that tocopherol and tocopheryl acetate are used in equal amounts, for example, 0.25% (w/w) tocopherol in combination with 0.25% (w/w) tocopheryl acetate, or 0.5% (w/w) tocopherol in combination with 0.5% (w/w) tocopheryl acetate.

(60) Other Cosmetic Excipients

(61) The skin care compositions described herein may include, apart from the above components, commonly known excipients, including perfumes, pigments, colorants, dyes, waxes, masking agents, humectants, surfactants, lubricants, stabilizers, sunscreens, emulsifiers, *medica*-ments, antiseptics, chelating agents, protectants, viscosifiers, vitamins, panthenol, ubiquin-one Q10, hyaluronic acid, or any combinations thereof.

(62) Bacterial Strains

(63) The skin care composition of the present invention comprises lyophilized or spray-dried live bacteria of at least one strain of the species *C. acnes*. It will however be preferred that the skin care composition comprises two or more strains of the species *C. acnes*. For example, the skin care composition may comprise 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more

than 20 strains of *C. acnes*. In some embodiments, the skin care composition comprises 2, 3, 4, or 5 different strains of *C. acnes*. It was reported that the reactivation and growth of some *C. acnes* strains is significantly supported when grown in a mixture of strains than when grown individually. For example, SLST type strain K8 grows slowly individually, but significantly faster when grown within a mixture of strains. Therefore, it is preferred according to the present invention that the composition comprises two or more strains of *C. acnes*.

(64) The bacterial species *Cutibacterium acnes* (*C. acnes*) was formerly known as *Propionibacterium acnes* (*P. acnes*). Based on the results from biochemical and genomic studies, the species was taxonomically reclassified in 2016. *C. acnes* is a Gram-positive, anaerobic, rod-shaped bacterium which is known to be involved in the development of acne and other pathological conditions. *C. acnes* strains occur on the skin of most people. *C. acnes* strains can be pathogenic or non-pathogenic. As used herein, “pathogenic” *C. acnes* strains are strains that are associated with acne. Assays for the identification and selection of pathogenic and non-pathogenic *C. acnes* strains are described in WO 2018/073651.

(65) *C. acnes* has been shown to comprise several distinct, major phylogenetic groups classified as types I, II and III, with the major type I clade being further divided into sub-clades known as types IA, IB and IC (Lomholt and Kilian, 2010). Sub-clade IA has been further subdivided into IA1 and IA2 (McDowell et al., 2012). Preferably, the at least one strain of the species *C. acnes* is a non-pathogenic strain of *C. acnes*. A genetic analysis of *C. acnes* strains revealed that strains which are non-pathogenic and not associated with acne are mainly members of (i) clade I, sub-clade IA2, (ii) clade I, sub-clade IB and (ii) clade II.

(66) Accordingly, in one embodiment of the invention, the at least one strain of the species *C. acnes* belongs either to one of sub-clades IA2, IB of clade I or to clade II. In one embodiment, the at least one strain of the species *C. acnes* belongs to sub-clade IA2. In another embodiment, the at least one strain of the species *C. acnes* belongs to sub-clade IB. In yet another embodiment, the at least one strain of the species *C. acnes* belongs to clade II. If more than one strain is used in the skin care composition of the present invention, it is preferred that strains from different clades or sub-clades are mixed with each other. For example, in one embodiment, the skin care composition comprises at least one strain from sub-clade IA2 and at least one strain from sub-clade IB. In another embodiment, the skin care composition comprises at least one strain from sub-clade IA2 and at least one strain from clade II. In yet another embodiment, the skin care composition comprises at least one strain from sub-clade IB and at least one strain from clade II.

(67) In other embodiments, the skin care composition of the present invention may comprise a mixture of *C. acnes* strains that include one or more clade I strains and one or more clade II strains. While clade II strains, as indicated above, may be less pathogenic than clade I strains, these strains can also be slower-growing than clade I strains, and less likely to be able to colonize the skin on their own. Accordingly, it may in some embodiments be advantageous that the skin care composition includes a mixture of strains that include both clade I and clade II strains which allow for comparatively improved colonization of the skin by clade II strains.

(68) Non-limiting examples for non-pathogenic strains of *C. acnes* include, but are not limited to, SLST type strains D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4. It is particularly preferred that the skin care composition of the present invention includes at least one SLST type C3 strain and/or at least one SLST type K8 strain. In one embodiment, the skin care composition of the present invention includes at least one SLST type C3 strain, and more preferably two or more SLST type C3 strains, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10 SLST type C3 strains. In another embodiment, the skin care composition of the present invention includes at least one SLST type K8 strain, and more preferably two or more SLST type K8 strains, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10 SLST type K8 strains. In yet another embodiment, the skin care composition of the present invention includes at least one SLST type C3 strain in combination with at least one SLST type K8 strain. In yet another embodiment, the skin care composition of the present

invention includes two or more SLST type C3 strains in combination with at least one SLST type K8 strain. In yet another embodiment, the skin care composition of the present invention includes at least one SLST type C3 strain in combination with two or more SLST type K8 strains. In yet another embodiment, the skin care composition of the present invention includes two or more SLST type C3 strains in combination with two or more SLST type K8 strains. In some embodiments, the one or more SLST type C3 strains and the one or more SLST type K8 strains are at approximately equal concentrations within the composition. In other embodiments, the one or more SLST type C3 strains are at a higher concentration than the one or more SLST type K8 strains within the composition. In other embodiments, the one or more SLST type C3 strains are at a lower concentration than the one or more SLST type K8 strains within the composition.

(69) In a particularly preferred embodiment, the skin care composition of the present invention includes the SLST type C3 strain that was deposited under the Budapest Treaty by S-Biomedic N. V. (Turnhoutsweg 30, 2340 Beerse, Belgium) at the Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures, Inhoffenstr. 7B, D-38124 Braunschweig, Germany on Oct. 19, 2017 under accession number DSM 32667. In another particularly preferred embodiment, the skin care composition of the present invention includes the SLST type K8 strain that was deposited under the Budapest Treaty by S-Biomedic N. V. (Turnhoutsweg 30, 2340 Beerse, Belgium) at the Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures, Inhoffenstr. 7B, D-38124 Braunschweig, Germany on Oct. 19, 2017 under accession number DSM 32668. In yet another preferred embodiment, the skin care composition of the present invention includes both the SLST type C3 strain DSM 32667 and the SLST type K8 strain DSM 32668. In another embodiment, the skin care composition comprising the at least one SLST type C3 strain and the at least one SLST type K8 strain additionally comprises at least one SLST type A5 strain and/or at least one SLST type F4 strain. Accordingly, in one embodiment, the composition comprises at least one SLST type C3 strain, at least one SLST type K8 strain, and at least one SLST type A5 strain. In another embodiment, the composition comprises at least one SLST type C3 strain, at least one SLST type K8 strain, and at least one SLST type F4 strain. In yet another embodiment, the composition comprises at least one SLST type C3 strain, at least one SLST type K8 strain, at least one SLST type A5 strain and at least one SLST type F4 strain.

(70) The strains designation referred to herein is based on the single-locus sequence typing (SLST) scheme described in Scholz et al. 2014 using locus PPA2385 as the SLST target sequence. The sequences of the PPA2385 locus of the different strains identified by Scholz are listed as SEQ ID NO:1-76 herein. Accordingly, a “SLST type C3” strain is a strain that comprises in its genome a sequence of the PPA2385 locus which is 100% identical to the sequence depicted in SEQ ID NO:27. Similarly, a “SLST type K8” strain is a strain that comprises in its genome a sequence of the PPA2385 locus which is 100% identical to the sequence as depicted in SEQ ID NO:64. The SLST scheme is also described in more detail in WO 2018/073651. Sequence identification of the PPA2385 locus can be performed as described in WO 2018/073651 by PCR amplification and DNA sequencing using the nucleotide primer set forth in SEQ ID NO:77-82. Based on the information presented herein and in WO 2018/073651, one of ordinary skill in the art would understand how *C. acnes* strains could be identified and classified.

(71) The composition the present invention may include both pathogenic strains and non-pathogenic strains of *C. acnes*. However, in a preferred embodiment of the invention, the skin care composition comprises exclusively non-pathogenic strains of *C. acnes*. It is particularly preferred that the skin care composition of the invention does not include a ribotype 6 (RT6) strain of *C. acnes*. The ribotype classification system is based on differences in the 16S rDNA sequence between different strains of *C. acnes*. The ribotype system is explained, for example, in Fitz-Gibbon et al. 2013. It is further particularly preferred that the skin care composition of the invention does not include a Phenotype III strain of *C. acnes*.

(72) *C. acnes* strains are normally able to produce the signaling molecule trans-10, cis-12 linoleic

acid from its precursor molecule linoleic acid, the latter of which is naturally present in the sebum (Rosson et al., 2004). Trans-10, cis-12 linoleic acid is thought to stimulate sebum production and secretion, which is important for *C. acnes* colonization of the skin. In this way, trans-10, cis-12 linoleic acid promotes the onset of acne (Downing et al., 1986; Letawe et al., 1998). Dependent on the skin of a subject, it may be favorable to either reduce or increase sebum production. For example, it may be useful to reduce sebum production in skin of a subject suffering from acne or oily skin. To the contrary, it may be useful to increase sebum production in skin of a subject suffering from dry skin.

(73) In one embodiment, the one or more *C. acnes* strains to be included into the skin care compositions of the present invention are hence selected based on its ability to produce trans-10, cis-12 linoleic acid. In one preferred embodiment, strains that produce low levels of trans-10, cis-12 linoleic acid are selected for inclusion into the skin care compositions of the invention which are to be used against acne or oily skin. Without wishing to be bound by theory, these strains are thought to reduce sebum production, which is useful for preventing or reducing the symptoms of acne or oily skin. SLST type strains C3, C1, F4, A5, K1, K2, K8 and L1 produce only low amounts of trans-10, cis-12 linoleic acid. In another preferred embodiment, strains that produce high levels of trans-10, cis-12 linoleic acid are selected for inclusion into the skin care compositions of the invention which are to be used against dry skin. Without wishing to be bound by theory, these strains are thought to increase sebum production, which is useful for preventing or reducing the symptoms of dry skin. SLST type strain A1 produces high amounts of trans-10, cis-12 linoleic acid.

(74) In one embodiment, the one or more *C. acnes* strains to be included into the skin care compositions of the present invention have been isolated from the skin microbiome of a donor subject. The subject may not be afflicted with acne or oily skin or may suffer from mild, moderate or severe acne. In another embodiment, the strains that have been isolated from the skin microbiome of a donor subject are non-pathogenic strains.

(75) The skin care composition of the present invention may also comprise one or more genetically modified strains of *C. acnes*. In another embodiment, the skin care composition comprises one or more genetically modified of *C. acnes* in combination with one or more naturally occurring strains of *C. acnes*. The genetically modified strains have preferably been modified to produce lower or higher amounts of trans-10, cis-12 linoleic acid. The production of trans-10, cis-12 linoleic acid can be detected as described in U.S. Pat. No. 6,743,609 or by other commonly known methods, such as FAME (fatty acid methyl esters) or gas chromatography. In certain other embodiments, the skin care composition of the present invention only comprises naturally occurring strains of *C. acnes*, i.e., only comprises strain(s) of *C. acnes* that have not been genetically modified by man.

(76) It is particularly preferred that the at least one *C. acnes* strain is present in the composition in an amount of 1.0×10^4 - 1.0×10^{11} colony forming units (CFU) per ml, more preferably 1.0×10^5 - 1.0×10^{10} CFU/ml, and even more preferably 1.0×10^7 - 1.0×10^{10} CFU/ml, or 1.0×10^8 - 1.0×10^{10} CFU/ml. For example, the at least one *C. acnes* strain may be present in an amount of at least 1.0×10^5 CFU/ml, preferably at least 1.0×10^6 CFU/ml, more preferably at least 1.0×10^7 CFU/ml, such as at least 1.0×10^8 CFU/ml, at least 1.0×10^9 CFU/ml, or at least 1.0×10^{10} CFU/ml of the skin care composition. It is particularly preferred that the at least one *C. acnes* strain is present in an amount of at least 1.0×10^{10} CFU/ml, 2.0×10^{10} CFU/ml, 3.0×10^{10} CFU/ml, 4.0×10^{10} CFU/ml, 5.0×10^{10} CFU/ml, 6.0×10^{10} CFU/ml, 7.0×10^{10} CFU/ml, 8.0×10^{10} CFU/ml, or 9.0×10^{10} CFU/ml of the skin care composition.

(77) In some embodiments, each of the *C. acnes* strains that is present in the composition is present in an amount of 1.0×10^4 - 1.0×10^{11} CFU/ml, more preferably 1.0×10^5 - 1.0×10^{10} CFU/ml, and even more preferably 1.0×10^7 - 1.0×10^{10} CFU/ml, or 1.0×10^8 - 1.0×10^{10} CFU/ml. For example, each of the *C. acnes* strains may be present in an amount of at least 1.0×10^5 CFU/ml, preferably at least 1.0×10^6 CFU/ml, more preferably at least

1.0×10⁷ CFU/ml, such as at least 1.0×10⁸ CFU/ml, at least 1.0×10⁹ CFU/ml, or at least 1.0×10¹⁰ CFU/ml of the skin care composition. It is particularly preferred that each of the *C. acnes* strains is present in an amount of at least 1.0×10¹⁰ CFU/ml, 2.0×10¹⁰ CFU/ml, 3.0×10¹⁰ CFU/ml, 4.0×10¹⁰ CFU/ml, 5.0×10¹⁰ CFU/ml, 6.0×10¹⁰ CFU/ml, 7.0×10¹⁰ CFU/ml, 8.0×10¹⁰ CFU/ml, or 9.0×10¹⁰ CFU/ml of the skin care composition. For example, if the skin care composition of the present invention comprises one SLST type C3 strain and one SLST type K8 strain, each of these strains may be present in an amount of 1.0×10⁴-1.0×10¹¹ CFU/ml, such as 1.0×10¹⁰ CFU/ml, 2.0×10¹⁰ CFU/ml, 3.0×10¹⁰ CFU/ml, 4.0×10¹⁰ CFU/ml, 5.0×10¹⁰ CFU/ml, 6.0×10¹⁰ CFU/ml, 7.0×10¹⁰ CFU/ml, 8.0×10¹⁰ CFU/ml, or 9.0×10¹⁰ CFU/ml.

(78) In some embodiments, the overall amount of lyophilized or spray-dried bacteria in the composition is 1.0×10⁴-1.0×10¹¹ CFU/ml, more preferably 1.0×10⁵-1.0×10¹⁰ CFU/ml, and even more preferably 1.0×10⁷-1.0×10¹⁰ CFU/ml, or 1.0×10⁸-1.0×10¹⁰ CFU/ml. For example, the bacteria may be collectively present in the composition in an amount of at least 1.0×10⁵ CFU/ml, preferably at least 1.0×10⁶ CFU/ml, more preferably at least 1.0×10⁷ CFU/ml, such as at least 1.0×10⁸ CFU/ml, at least 1.0×10⁹ CFU/ml, or at least 1.0×10¹⁰ CFU/ml of the skin care composition. It is particularly preferred that the bacteria are collectively present in the composition in an amount of at least 1.0×10¹⁰ CFU/ml, 2.0×10¹⁰ CFU/ml, 3.0×10¹⁰ CFU/ml, 4.0×10¹⁰ CFU/ml, 5.0×10¹⁰ CFU/ml, 6.0×10¹⁰ CFU/ml, 7.0×10¹⁰ CFU/ml, 8.0×10¹⁰ CFU/ml, or 9.0×10¹⁰ CFU/ml of the skin care composition. One of ordinary skill in the art will be readily able to determine the amount of bacteria in a lyophilized or spray-dried composition.

(79) Preferred Compositions

(80) The above components described as being suitable for use in the skin care composition of the present invention can be combined with each other without limitation. Accordingly, skin care compositions of the inventions may comprise, apart from the lyophilized or spray-dried live bacteria of at least one *C. acnes* strain and the thickener, at least one excipient selected from the group consisting of emollients, fillers, antioxidants, preservatives, solubilizers, and pH adjusters, wherein said one or more excipients do not interfere with viability and reactivation of growth of said lyophilized or spray-dried live bacteria. Particularly preferred skin care compositions covered by the instant disclosure are described in the following.

(81) Another Preferred Skincare Composition Comprises

(82) (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain; and (b) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain; and (b) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain; and (b) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain; and (b) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain selected from the group consisting of SLST type strains D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4; and (b) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain selected from the group consisting of SLST type strains D1, A5, C1, C3,

H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4; and (b) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain selected from the group consisting of SLST type strains D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4; and (b) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain selected from the group consisting of SLST type strains D1, A5, C1, C3, H1, H2, H3, K1, K2, K4, K6, K8, K9, L1, and F4; and (b) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; and (b) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; and (b) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; and (b) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; and (b) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) dicaprylyl carbonate, and preferably 7.5 to 10.0% (w/w) dicaprylyl carbonate; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) ethylhexyl cocoate, and preferably 7.5 to 10.0% (w/w) ethylhexyl cocoate; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) dicaprylyl carbonate, and preferably 7.5 to 10.0% (w/w) dicaprylyl carbonate; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) ethylhexyl cocoate, and preferably 7.5 to 10.0% (w/w) ethylhexyl cocoate; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) dicaprylyl carbonate, and preferably 7.5 to 10.0% (w/w) dicaprylyl carbonate; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w)

ethylhexyl cocoate, and preferably 7.5 to 10.0% (w/w) ethylhexyl cocoate; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) dicaprylyl carbonate, and preferably 7.5 to 10.0% (w/w) dicaprylyl carbonate; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 25.0% (w/w) ethylhexyl cocoate, and preferably 7.5 to 10.0% (w/w) ethylhexyl cocoate; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 7.5% (w/w) of a *C. crispus* extract, and preferably 0.2 to 1.5% (w/w) of a *C. crispus* extract; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 10.0% (w/w) hydroxypropyl starch phosphate, and preferably 2.0 to 4.0% (w/w) hydroxypropyl starch phosphate; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 7.5% (w/w) of a *C. crispus* extract, and preferably 0.2 or 1.5% (w/w) of a *C. crispus* extract; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 10.0% (w/w) hydroxypropyl starch phosphate, and preferably 2.0 to 4.0% (w/w) hydroxypropyl starch phosphate; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 7.5% (w/w) of a *C. crispus* extract, and preferably 0.2 or 1.5% (w/w) of a *C. crispus* extract; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 10.0% (w/w) hydroxypropyl starch phosphate, and preferably 2.0 to 4.0% (w/w) hydroxypropyl starch phosphate; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 7.5% (w/w) of a *C. crispus* extract, and preferably 0.2 or 1.5% (w/w) of a *C. crispus* extract; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of

at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 10.0% (w/w) hydroxypropyl starch phosphate, and preferably 2.0 to 4.0% (w/w) hydroxypropyl starch phosphate; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol; and (c) a citric acid/citrate buffer that provides for a pH of the composition of about 4.0 to about 7.0.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol; and (c) a citric acid/citrate buffer that provides for a pH of the composition of about 4.0 to about 7.0.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol; and (c) a citric acid/citrate buffer that provides for a pH of the composition of about 4.0 to about 7.0.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol; and (c) a citric acid/citrate buffer that provides for a pH of the composition of about 4.0 to about 7.0.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) distarch phosphate, and preferably 2.0 to 3.0% (w/w) distarch phosphate; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) tapioca starch, and preferably 2.0 to 3.0% (w/w) tapioca starch; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) distarch phosphate, and preferably 2.0 to 3.0% (w/w) distarch phosphate; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) tapioca starch, and preferably 2.0 to 3.0% (w/w) tapioca starch; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) distarch phosphate, and preferably 2.0 to 3.0% (w/w) distarch phosphate; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) tapioca starch, and preferably 2.0 to 3.0% (w/w) tapioca starch; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) distarch phosphate, and preferably 2.0 to 3.0% (w/w) distarch phosphate; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.05 to 5.0% (w/w) tapioca starch, and preferably 2.0 to 3.0% (w/w) tapioca starch; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.5% (w/w) PEG-40 hydrogenated castor oil, and preferably 0.5 to 1.0% (w/w) PEG-40 hydrogenated castor oil; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.5% (w/w) PEG-40 hydrogenated castor oil, and preferably 0.5 to 1.0% (w/w) PEG-40 hydrogenated castor oil; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.5% (w/w) PEG-40 hydrogenated castor oil, and preferably 0.5 to 1.0% (w/w) PEG-40 hydrogenated castor oil; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.5% (w/w) PEG-40 hydrogenated castor oil, and preferably 0.5 to 1.0% (w/w) PEG-40 hydrogenated castor oil; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopherol, and preferably 0.1 to 1.0% (w/w) tocopherol; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopherol, and preferably 0.1 to 1.0% (w/w) tocopherol; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopherol, and preferably 0.1 to 1.0% (w/w) tocopherol; and (c) 0.05 to 0.5% (w/w) caprylyl

glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopherol, and preferably 0.1 to 1.0% (w/w) tocopherol; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopheryl acetate, and preferably 0.1 to 1.0% (w/w) tocopheryl acetate; and (c) 0.5 to 20.0% (w/w) ethanol, and preferably 8.0 to 15.0% (w/w) ethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopheryl acetate, and preferably 0.1 to 1.0% (w/w) tocopheryl acetate; and (c) 0.05 to 0.5% (w/w) phenoxyethanol, and preferably 0.1 to 0.2% (w/w) phenoxyethanol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopheryl acetate, and preferably 0.1 to 1.0% (w/w) tocopheryl acetate; and (c) 0.05 to 0.5% (w/w) caprylyl glycol, and preferably 0.1 to 0.2% (w/w) caprylyl glycol.

Another Preferred Skincare Composition Comprises (a) lyophilized or spray-dried live bacteria of at least one *C. acnes* strain, preferably lyophilized or spray-dried live bacteria of at least one *C. acnes* SLST type C3 strain and at least one *C. acnes* SLST type K8 strain; (b) 0.01 to 2.0% (w/w) tocopheryl acetate, and preferably 0.1 to 1.0% (w/w) tocopheryl acetate; and (c) 0.05 to 5.0% (w/w) methylpropanediol, and preferably 1.0 to 5.0% (w/w) methylpropanediol.

Cosmetic and Therapeutic Uses

(83) The skin care composition described hereinabove are useful for the modulation of the skin microbiome, and in particular for maintaining healthy skin, such as skin that is free of acne.

Compositions comprising at least one live *C. acnes* bacterial strain can help skin to revert microbiome disease states to healthy microbiome states. In one preferred embodiment, the skin care composition is used for preventing the formation of acne or treating acne. In another preferred embodiment, the skin care composition is used for preventing the reoccurrence of acne in a subject who has received a standard acne treatment. It is particularly preferred that the subject is a human.

(84) In one aspect, the invention provides a skin care composition as described hereinabove for use in a method of improving the appearance of the skin of a subject and/or modulating (e.g. increasing or decreasing) the sebum production of skin cells of a subject and/or maintaining healthy skin, such as skin that is free of acne, of a subject. In yet another aspect, the invention relates to the use of a skin care composition as described hereinabove for improving the appearance of the skin of a subject and/or for modulating (e.g. increasing or decreasing) the sebum production of skin cells of a subject and/or for maintaining healthy skin, such as skin that is free of acne, of a subject. In one aspect, the invention provides a skin care composition as described hereinabove for use in a method of treating or preventing a condition selected from the group consisting of acne, oily skin, progressive macular hypomelanosis, dandruff, atopic eczema, atopic dermatitis and rosacea in a subject. It is particularly preferred that the subject is a human.

(85) Methods for treating the skin of a subject by administering a skin care composition as described hereinabove are also provided. These methods may be cosmetic or therapeutic methods. In one aspect a method of improving the appearance of the skin of a subject and/or modulating the sebum production of skin cells of a subject and/or maintaining healthy skin of a subject is provided, said method comprising the topical administration of a skin care composition as described

hereinabove. In another aspect, the invention provides a method of treating or preventing a condition selected from the group consisting of acne, oily skin, progressive macular hypomelanosis, dandruff, atopic eczema, atopic dermatitis and rosacea in a subject, said method comprising the topical administration of a skin care composition described hereinabove. It is particularly preferred that the subject is a human.

(86) When the compositions are applied to the skin, it is preferred that the amount of the composition applied to the skin is between 0.5 g and 2.0 g, more preferably between 0.5 g and 1.0 g. Stated differently, the amount of the composition may correspond to at least $1.0 \times 10^{5.5}$ CFU, at least $1.0 \times 10^{6.6}$ CFU, at least $1.0 \times 10^{7.7}$ CFU, at least $1.5 \times 10^{7.7}$ CFU, at least $2.0 \times 10^{7.7}$ CFU, or at least $2.5 \times 10^{7.7}$ CFU.

(87) Ready-to-Use Compositions and Kit-of-Parts

(88) The skin care composition of the present invention may be provided as ready-to-use composition which is suitable for direct topical administration to the skin. Such a composition may be provided in different forms, including, but not limited to, in the form of a gel, cream, lotion, ointment, paste, soft paste, suspension, solution, salve, wax, milk, emulsion, or the like. In such a composition, the lyophilized or spray-dried live bacteria will be present in admixture with other cosmetic or pharmaceutical excipients described elsewhere herein, such as emollients, fillers, and the like. Upon application of these compositions to the skin, the dried bacteria will be re-activated on the skin of the subject to which the product is applied. Growth of the re-activated bacteria from the skin care composition will positively influence the microbial flora on the skin of the subject.

(89) These ready-to-use compositions are preferably stable at room temperature for at least 1 week, at least 2 weeks, at least 3 weeks, at least 4 weeks, at least 5 weeks, at least 6 weeks, at least 7 weeks, at least 8 weeks, at least 9 weeks, at least 10 weeks, at least 11 weeks, at least 12 weeks, at least 13 weeks, at least 14 weeks, at least 15 weeks, at least 16 weeks, at least 17 weeks, at least 18 weeks, at least 19 weeks, at least 20 weeks, at least 21 weeks, at least 22 weeks, at least 23 weeks, at least 24 weeks, at least 25 weeks, at least 26 weeks, at least 27 weeks, at least 28 weeks, at least 29 weeks, at least 30 weeks or more than 30 weeks. Stated differently, such compositions are preferably stable at room temperature for at least 1 month, at least 2 months, at least 3 months, at least 4 months, at least 5 months, at least 6 months or more than 6 months. As used herein, a composition is regarded as being stable if the reduction in the number of colony forming units present in the composition after storage is less than a 3 log reduction, preferably less than a 2 log reduction, and more preferably less than a 1 log reduction. Stated differently, a composition is regarded as being stable if the reduction in the number of colony forming units present in the composition after storage is less than 1000-fold, preferably less than 100-fold, and more preferably less 10-fold relative to the number of colony forming units in the composition before storage.

(90) The skin care compositions of the present invention may alternatively be provided as a kit-of-parts in which the lyophilized or spray-dried bacteria are spatially separated from the other components, e.g. the cosmetic or therapeutic components. For example, the kit-of-parts may be in the form of a packaging with two spatially separated chambers, wherein the first chamber contains the lyophilized or spray-dried bacteria, and the second chamber contains a cosmetic preparation, such as a water-containing cosmetic preparation. Prior to use, the contents of both chambers are mixed with each other, such as for example by a consumer or a patient, to provide a homogeneous skin care composition which is then applied to the skin. A kit-of-part assembly has the advantage that the bacteria can remain in lyophilized or spray-dried form until use which is associated with a particular high storage stability of the composition. In a kit-of-parts assembly, it is advantageous if the weight ratio of the bacteria in the first chamber to the cosmetic preparation, in particular the water-containing cosmetic preparation, in the second chamber is from 1:10 to 1:100, such as 1:10, 1:20, 1:30, 1:40, 1:50, 1:60, 1:70, 1:80, 1:90, or 1:100. After mixing the contents of both chambers, the skin care composition contains preferably 1-10% by weight lyophilized or spray-dried bacteria and 99-90% by weight of the cosmetic preparation, e.g. the water-containing cosmetic preparation.

According to the present invention a kit-of-parts can be provided, for example, in a Lyo-Ject® double-chamber syringe, in a V-LK® double-chamber carpuel or in a dual-chamber system as described in WO2018077598 A1. In another preferred embodiment, the lyophilized or spray-dried bacteria in the first chamber may be suspended in a lipid or oil. This will significantly facilitate packaging and filling. In addition, the surrounding lipid or oil will protect the bacteria from premature rehydration. Preferably, the bacteria are suspended in ethylhexyl cocoate or dicaprylyl carbonate. The weight ratio of the bacteria to the oil or lipid preferably is between 1:1 and 1:2.

(91) In one embodiment, the skin care composition of the present invention is an aqueous preparation, such as a gel. Aqueous preparations as intended herein encompass aqueous solutions, as well as aqueous dispersions. In one embodiment, the skin care composition is an oil-in-water emulsion. If the skin care composition contains an oil phase, e.g. when using an oil-in-water emulsion, it is preferred that the oil phase contains triglycerides and/or octyldodecanol. In addition, the oil phase may contain one or more oils selected from the group of lecithin, olive oil, sunflower oil, jojoba oil, soya oil, peanut oil, rapeseed oil, almond oil, palm oil, coconut oil, castor oil, wheat germ oil, grape seed oil, safflower oil, evening primrose oil, macadamia nut oil and the like.

(92) The term “comprising”, when used in the context with methods or compositions, means that other method steps or components of the composition can be present in addition to the method steps or components presented. The use of the term “comprising” indicates inclusion rather than limitation. For example, a composition “comprising” components A+B may also comprise C as a further component. Similarly, a method “comprising” steps (a) and (b) may also comprise (c) as a further method step. In contrast, the term “consisting of”, when used in the context with methods or compositions, refers to methods or compositions which are exclusive of any other method steps or components of the composition not recited in the description of the respective composition or method. For example, a composition “consisting of” components A+B is limited to these two components and does not contain any other component apart from A and B. Similarly, a method “consisting of” steps (a) and (b) is a two-step method and does not contain any other method steps apart from (a) and (b). It should be understood, however, that any method or composition described herein as “comprising” certain method steps or components may preferably “consist essentially of” or may more preferably “consist of” the recited method steps or components. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.

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DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE INVENTION

Examples

(94) The present invention is further illustrated by the following examples, which in no way should be construed as further limiting. The entire contents of all of the references (including literature references, issued patents, published patent applications, and co pending patent applications) cited throughout this application are hereby expressly incorporated by reference.

Example 1: Evaluation of the Minimum Inhibitory Concentration (MIC)

(95) In microbiology, the MIC is the lowest concentration of a chemical which prevents visible growth of bacterium. The resulted MIC value depends both on the microorganism and the test compound. The MIC is determined by preparing concentrations series of the test compound in vitro, incubating the solutions with the separate batches of cultured bacteria, and measuring the results using agar dilution or broth microdilution. For the experiments with *C. acnes* strains C3 and K8, media comprising yeast extract soy peptone and dextrose were used. The MIC is determined by turbidity.

(96) A total of 125 cosmetic excipients were tested for their compatibility with *C. acnes* strains C3 and K8. Hydrophilic compounds were tested in a 96-well plate liquid broth format, poorly soluble compounds were pre-dissolved in DMSO before dilution in liquid broth and hydrophobic or solid compounds were tested in an agar well diffusion test assay. The different dilutions of the test excipients in media were prepared in Falcon tubes and then distributed in a deep well plate using a multistepper. Each well received 1.2 ml and was subsequently inoculated with 120 µl of *C. acnes* bacteria suspension (strain C3 and K8). From the deep well plate 200 µl were transferred in each well of a clear 96 well replicate plate. The leftover was discarded. The plates were then incubated anaerobically for 3 days at 37° C. Following incubation, the test plates were visually examined and wells were scored for growth or complete growth inhibition to define the minimum inhibitory concentration.

(97) Results:

(98) It was found that the compatibility of standard excipients for the bacterial strains widely varied. Only few excipients did not lead to inhibition or significant inhibition. These excipients are listed in the below table with their compatibility values for *C. acnes* strains C3 and K8. The compatibility values represent the MIC value of the listed ingredients with the tested *C. acnes* strains. At this concentration [% w/vol] the tested compound starts interfering with the growth of the tested strains. The compounds can be used in formulations at concentrations below this concentration.

(99) The results obtained from MIC testing are depicted in the below Table 1.

(100) TABLE-US-00001 COMPATABILITY COMPATABILITY INCI FUNCTION VALUE C3 [%] VALUE K8 [%]

dicaprylyl carbonate + emollient	10.00	10.00	tocopherol ethylhexyl cocoate		
emollient	10.00	10.00	<i>Chondrus crispus</i> extract thickener	5.00	5.00
hydroxypropyl starch thickener	1.00	1.00	phosphate + aqua distarch phosphate filler	5.00	5.00
tapioca starch + aqua filler	5.00	5.00	tocopherol antioxidant	1.50	1.50
tocopheryl acetate antioxidant	1.00	1.00	ethanol preservative	3.75	3.75
phenoxyethanol preservative	0.45	0.23	caprylyl glycol preservative	0.18	0.18
methylpropanediol preservative	4.00	4.00	peg-40 hydrogenated solubilizer	1.00	1.00
castor oil			citric acid buffering	0.33	0.08
Citric acid/citrate buffering	1.30	1.30			

Example 2: Reactivation Assay of Freeze Dried Bacteria in Prototype Formulations

(101) Based on the results from the MIC testing, a set of 15 prototype formulations were designed to test for efficient reactivation of freeze-dried *C. acnes* bacteria. For this purpose, a freeze-dried powder of *C. acnes* strains K8 and C3 was produced and mixed with other ingredients as indicated below. Ingredients are listed in % (w/w). The formulations were provided in two pre-mixed compositions A and B that were finally mixed with each other. The following formulations were made:

(102) TABLE-US-00002 Formulation 1 A Isopropyl Palmitate 20.000 Carrageenan 1.000 Sodium chloride 0.900 Sodium citrate 0.174 Citric acid 0.015 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Water ad 100

(103) TABLE-US-00003 Formulation 2 A Isopropyl Palmitate 20.000 Carrageenan 1.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium chloride 0.668 Sodium citrate 0.129 Citric acid 0.064 Glycerin 5.565 Water ad 100

(104) TABLE-US-00004 Formulation 3 A Isopropyl Palmitate 20.000 Carrageenan 1.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium chloride 0.668 Sodium citrate 0.129 Citric acid 0.015 Glycerin 5.565 Ethanol 8.040 Water ad 100

(105) TABLE-US-00005 Formulation 4 A Isopropyl Palmitate 20.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Carrageenan 0.075 Sodium chloride 0.677 Sodium citrate 0.131 Citric acid 0.015 Glycerin 5.640 Water ad 100

(106) TABLE-US-00006 Formulation 5 A Isopropyl Palmitate 20.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Hydroxyethyl cellulose 0.451 Sodium chloride 0.677 Sodium citrate 0.131 Citric acid 0.015 Glycerin 5.640 Ethanol 8.152 Water ad 100

(107) TABLE-US-00007 Formulation 6 A Isopropyl Palmitate 20.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Hydroxypropyl starch phosphate 3.384 Sodium chloride 0.677 Sodium citrate 0.131 Citric acid 0.008 Glycerin 5.640 Ethanol 8.152 Water ad 100

(108) TABLE-US-00008 Formulation 7 A Isopropyl Palmitate 20.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium hyaluronate 0.376 Carbomer 0.564 Sodium chloride 0.677 Sodium citrate 0.131 Citric acid 0.008 Glycerin 3.760 Ethanol 8.152 Caprylyl Glycol 0.075 Water ad 100

(109) TABLE-US-00009 Formulation 8 A Isopropyl Palmitate 20.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium hyaluronate 0.978 Sodium chloride 0.677 Sodium citrate 0.131 Citric acid 0.008 Glycerin 3.760 Ethanol 8.152 Caprylyl Glycol 0.075 Butylene glycol 3.760 Water ad 100

(110) TABLE-US-00010 Formulation 9 A Carrageenan 1.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium chloride 0.848 Sodium citrate 0.164 Citric acid 0.019 Glycerin 7.065 Ethanol 10.211 Water ad 100

(111) TABLE-US-00011 Formulation 10 A Carrageenan 1.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 2.800 B Sodium chloride 0.848 Sodium citrate 0.164 Citric acid 0.019 Glycerin 7.065 Phenoxyethanol 0.188 Methylpropanediol 3.768 Caprylyl Glycol 0.094 Water ad 100

(112) TABLE-US-00012 Formulation 11 A *C. acnes* lyophilisate 2.800 B Hydroxyethyl cellulose 0.583 Sodium chloride 0.875 Sodium citrate 0.169 Citric acid 0.019 Glycerin 7.290 Ethanol 10.536 Water ad 100

(113) TABLE-US-00013 Formulation 12 A Isopropyl Palmitate 5.000 Ethylhexyl cocoate 10.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 0.940 B Hydroxyethyl cellulose 0.498 Sodium chloride 0.747 Sodium citrate 0.144 Citric acid 0.017 Glycerin 6.225 Ethanol 9.740 Water ad 100

(114) TABLE-US-00014 Formulation 13 A Dicaprylyl Carbonate 7.500 Ethylhexyl cocoate 7.500 Distarch phosphate 2.000 *C. acnes* lyophilisate 0.940 B Hydroxyethyl cellulose 0.498 Sodium chloride 0.747 Sodium citrate 0.144 Citric acid 0.017 Glycerin 6.225 Ethanol 9.740 Water ad 100

(115) TABLE-US-00015 Formulation 14 A Isodecyl neopentanoate 10.000 Isopropyl palmitate 5.000 Distarch phosphate 2.000 *C. acnes* lyophilisate 0.940 B Hydroxyethyl cellulose 0.498

Sodium chloride 0.747 Sodium citrate 0.144 Citric acid 0.017 Glycerin 6.225 Ethanol 9.740 Water ad 100

(116) TABLE-US-00016 Formulation 15 A Dicaprylyl ether 7.500 Isopropyl palmitate 7.500 Distarch phosphate 2.000 *C. acnes* lyophilisate 0.940 B Hydroxyethyl cellulose 0.498 Sodium chloride 0.747 Sodium citrate 0.144 Citric acid 0.017 Glycerin 6.225 Ethanol 9.740 Water ad 100

(117) The formulations are depicted in the form of a table in FIG. 1. For formulations 1-11, the CFU target value was 1×10^7 . For formulations 12-15, the CFU target value was 1×10^6 . A CFU count was performed directly after mixing the formulation to assess the effect of the prototype formulation on the revival and subsequent stability rate of the bacteria. Briefly, the formula component A (lyophilisate/oil suspension) and the formula component B (hydrogel/aqueous solution) are mixed together and well vortexed. Directly after mixing an aliquot is removed and transferred to 0.9% NaCl. Further dilutions are plated on COST agar plates and incubated anaerobically at 37° C. for 5 days.

(118) Results:

(119) The results are depicted in FIGS. 2 and 3. It can be seen that all formulations 1-15 showed an acceptable reactivation of the freeze-dried bacteria. There was no negative influence of the ingredients on bacteria reactivation.

Example 3: Reactivation Assay of Freeze Dried Bacteria in Prototype Formulations

(120) Similar to the formulations tested in Example 2, another set of 11 prototype formulations was prepared with higher concentrations of the excipients of the invention. The formulations were tested as described in Example 2 for their impact on reactivation of freeze-dried *C. acnes* bacteria. As a control, a formulation containing the lyophilisate and low amounts of the excipients was prepared. The following formulations were made:

(121) TABLE-US-00017 Control A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(122) TABLE-US-00018 Formulation 1 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 25.000 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(123) TABLE-US-00019 Formulation 2 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 5.000 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(124) TABLE-US-00020 Formulation 3 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 20.000 Water ad 100

(125) TABLE-US-00021 Formulation 4 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 2.500 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(126) TABLE-US-00022 Formulation 5 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Dicaprylyl Carbonate 25.000 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(127) TABLE-US-00023 Formulation 6 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001

Hydroxyethylcellulose 0.050 Glycerin 0.050 Chondrus Crispus Extract 7.500 Ethanol 0.500 Water ad 100

(128) TABLE-US-00024 Formulation 7 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Hydroxypropylstarch Phosphate 10.000 Ethanol 0.500 Water ad 100

(129) TABLE-US-00025 Formulation 8 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Tocopherol 2.000 Ethanol 0.500 Water ad 100

(130) TABLE-US-00026 Formulation 9 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Tocopheryl Acetate 2.000 Ethanol 0.500 Water ad 100

(131) TABLE-US-00027 Formulation 10 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 Tapioca Starch 5.000 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Glycerin 0.050 Ethanol 0.500 Water ad 100

(132) TABLE-US-00028 Formulation 11 A Isopropyl Palmitate 0.050 Ethylhexyl Cocoate 0.050 Distarch phosphate 0.050 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.010 Sodium Chloride 0.050 Sodium Citrate 0.010 Citric Acid 0.010 Sodium Hydroxide 0.001 Hydroxyethylcellulose 0.050 Phenoxyethanol 0.230 Caprylyl Glycol 0.180 Methylpropanediol 4.000 Glycerin 0.050 Water ad 100

(133) The formulations are depicted in the form of a table in FIG. 4. For all formulations, the CFU target value was 1×10^9 . A CFU count was performed directly after mixing the formulation, as described above in Example 2, and after 1 hour.

(134) Results:

(135) The results are depicted in FIG. 5. It can be seen that all formulations 1-11 showed an acceptable reactivation of the freeze-dried bacteria. There was no negative influence of the ingredients on bacteria reactivation even at the high concentrations that were used in this experiment.

Example 4: Comparative Reactivation Assays

(136) To compare the excipients identified herein as particularly useful for compositions containing lyophilized or spray-dried live bacteria with other excipients having the same or similar function, comparative formulations V-1 to V-10 were prepared and tested for their influence on bacteria reactivation as described in Example 2. As a positive control, a lyophilisate-containing formulation was used which had been found not to influence bacteria reactivation negatively. The following formulations were made for comparison:

(137) TABLE-US-00029 Control A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(138) TABLE-US-00030 Formulation V-1 A Paraffinum Liquidum 18.750 Distarch phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(139) TABLE-US-00031 Formulation V-2 A Octyldodecanol 18.750 Distarch phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin

5.553 Ethanol 9.625 Water ad 100

(140) TABLE-US-00032 Formulation V-3 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Talc 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(141) TABLE-US-00033 Formulation V-4 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Polymethylsilsesquioxane 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(142) TABLE-US-00034 Formulation V-5 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Carbomer 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(143) TABLE-US-00035 Formulation V-6 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Alkyl/Acrylate Crosspolymer 0.444 Glycerin 5.553 Ethanol 9.625 Water ad 100

(144) TABLE-US-00036 Formulation V-7 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Ascorbic Acid 3.000 Ethanol 9.625 Water ad 100

(145) TABLE-US-00037 Formulation V-8 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B PEG-40 Hydrogenated Castor Oil 0.370 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Benzethonium Chloride 0.090 Water ad 100

(146) TABLE-US-00038 Formulation V-9 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B Caprylyl/Capryl Glucoside 1.000 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Water ad 100

(147) TABLE-US-00039 Formulation V-10 A Isopropyl Palmitate 6.250 Ethylhexyl Cocoate 12.500 Distarch Phosphate 2.500 *C. acnes* lyophilisate 4.710 B Glyceryl Caprylate 0.300 Sodium Chloride 0.677 Sodium Citrate 0.129 Citric Acid 0.015 Sodium Hydroxide 0.013 Hydroxyethylcellulose 0.444 Glycerin 5.553 Water ad 100

(148) The formulations are depicted in the form of a table in FIG. 7. For all formulations, the CFU target value was 1×10^9 . A CFU count was performed directly after mixing the formulation, as described above in Example 2, and also after 1 hour or 6 hours, respectively.

(149) Results:

(150) The results are depicted in FIG. 6. It can be seen that formulations V-1 to V-10 have a clear negative impact on the reactivation of the bacteria. This demonstrates that numerous excipients commonly used in cosmetic preparations negatively interfere with the viability or reactivation of the freeze-dried bacteria used in the present studies.

Claims

1. A skin care composition, wherein the composition is suitable for topical application to skin and comprises: (a) lyophilized and/or spray dried live bacteria of at least one cutibacterium *acnes* (*C. acnes*) single-locus sequence typing (SLST) type C3 strain and/or at least one *C. acnes* SLST type K8 strain; and (b) one or more of from 0.05% to 0.5% (w/w) of phenoxyethanol, from 0.05% to 5.0% (w/w) of caprylyl glycol, from 0.05% to 5.0% (w/w) of methylpropanediol as preservative.
2. The skin care composition of claim 1, wherein the composition at least comprises from 0.05% to

- 0.5% (w/w) of phenoxyethanol.
3. The skin care composition of claim 1, wherein the composition at least comprises from 0.05% to 5.0% (w/w) of caprylyl glycol.
4. The skin care composition of claim 1, wherein the composition at least comprises from 0.05% to 5.0% (w/w) of methylpropanediol.
5. The skin care composition of claim 1, wherein the composition comprises at least all of phenoxyethanol, caprylyl glycol and methylpropanediol.
6. The skin care composition of claim 5, wherein the composition comprises from 0.1% to 0.3% (w/w) of phenoxyethanol, from 0.1% to 0.3% (w/w) of caprylyl glycol and from 2.0% to 5.0% (w/w) of methylpropanediol.
7. The skin care composition of claim 1, wherein the composition further comprises dicaprylyl carbonate and/or ethylhexyl cocoate as emollient.
8. The skin care composition of claim 1, wherein the composition further comprises *Chondrus crispus* extract and/or hydroxypropyl starch phosphate as thickener.
9. The skin care composition of claim 1, wherein the composition further comprises distarch phosphate and/or tapioca starch as filler.
10. The skin care composition of claim 1, wherein the composition further comprises one or both of tocopherol and tocopheryl acetate as antioxidant.
11. The skin care composition of claim 1, wherein the composition further comprises PEG-40 hydrogenated castor oil as solubilizer.
12. The skin care composition of claim 1, wherein the at least one strain of *C. acnes* further comprises at least one SLST type strain A5 and/or at least one SLST type strain F4.
13. The skin care composition of claim 1, wherein a concentration of each type of *C. acnes* strain present in the composition is at least 0.5% (w/v) of the skin care composition.
14. The skin care composition of claim 1, wherein each type of the *C. acnes* strain in the composition is present in an amount of from 10^{sup.4} to 10^{sup.11} CFU/ml.
15. The skin care composition of claim 1, wherein the composition is present in the form of at least one of a gel, a cream, an ointment or a lotion.
16. The skin care composition of claim 1, wherein the composition is capable of treating or preventing one or more of acne, oily skin, progressive macular hypomelanosis, dandruff, atopic eczema, atopic dermatitis, rosacea in a subject.
17. The skin care composition of claim 1, wherein the composition is capable of improving an appearance of skin of a subject and/or of modulating a sebum production of skin cells of a subject and/or of maintaining healthy skin of a subject.
18. The skin care composition of claim 1, wherein the composition comprises at least one SLST type strain C3.
19. The skin care composition of claim 1, wherein the composition comprises at least one SLST type strain K8.
20. The skin care composition of claim 1, wherein the composition comprises at least one SLST type strain D3 and at least one SLST type strain K8.
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